Modeling with partial differential equations

Exercises – Session 1: Ordinary Differential Equations

Exercise 1 (Duhamel/Variation of Constants for a Linear System)**.** *Consider the ordinary differential equation on* $X\colon [0,T] \to \mathbb{R}^d$*:*

$$\dot{X}(t) = A\,X(t) + B(t), \qquad X(0) = X_0, \tag{1}$$

where $A \in \mathbb{R}^{d\times d}$ *is constant and* $B : [0,T] \to \mathbb{R}^d$ *is continuous.*

(a) *Show that the solution of the homogeneous system* $\dot{X} = AX$ *with* $X(0) = X_0$ *is*

$$X_{\text{hom}}(t) = \mathrm{e}^{tA}\,X_0,$$

where $\mathrm{e}^{tA} := \sum_{k=0}^{\infty} \frac{t^k}{k!} A^k$*. (You may verify that* $\frac{d}{dt}\mathrm{e}^{tA} = A\mathrm{e}^{tA}$*.)*

Correction. *Since* $e^{tA} = \sum_{k\geq 0} \frac{t^k}{k!} A^k$*, we have*

$$\frac{d}{dt}e^{tA} = \sum_{k\geq 1} \frac{k\,t^{k-1}}{k!} A^k = \sum_{k\geq 1} \frac{t^{k-1}}{(k-1)!} A^k = A \sum_{j\geq 0} \frac{t^j}{j!} A^j = Ae^{tA}.$$

Notice that all the series are convergent since they are absolutely (or normally) convergent, for any submultiplicative matrix norm (see below for details). Thus, one has $\frac{d}{dt}\big(e^{tA}X(0)\big) = A\big(e^{tA}X(0)\big)$ *and* $e^{0\,A}X(0) = X(0)$*. The function* $t \mapsto e^{tA}X(0)$ *verifies the homogeneous equation and by unicity of the solution, it is the desired solution* X_{hom}*.*

About exponential of matrices. *Let us recall that* $\sum_{k=0}^{\infty} \frac{t^k}{k!} A^k$ *is well defined if* $\sum_{k=0}^{\infty} \frac{t^k}{k!} \|A^k\| < \infty$*. This is obvious for submultiplicative matrix norms, namely norms such that: for all* $A, B \in \mathbb{R}^{d\times d}$*,* $\|AB\| \leq \|A\|\|B\|$*. An example of such a norm is the Frobenius norm (namely the Euclidean norm for the matrix seen as a* d^2*-dimensional vector):*

$$\|A\|_{Fr} = \sqrt{\sum_{1\leq i,j\leq d} A_{i,j}^2}.$$

The Frobenius norm is submultiplicative by Cauchy Schwarz, since: for any $1 \leq i, j \leq d$*,*

$$\begin{aligned}
\Big|[AB]_{i,j}\Big| &= \left|\sum_{k=1}^{d} A_{i,k}B_{k,j}\right| \\
&\leq \sum_{k=1}^{d} |A_{i,k}|\,|B_{k,j}| \\
&\leq \sqrt{\sum_{k=1}^{d} |A_{i,k}|^2}\sqrt{\sum_{k=1}^{d} |B_{k,j}|^2}
\end{aligned}$$

and thus,

$$\sum_{1\leq i,j\leq d} \Big|[AB]_{i,j}\Big|^2 \leq \sum_{1\leq i,j\leq d} \left(\sum_{k=1}^{d} |A_{i,k}|^2 \sum_{k=1}^{d} |B_{k,j}|^2\right) = \left(\sum_{1\leq i,k\leq d} |A_{i,k}|^2\right)\left(\sum_{1\leq i,k\leq d} |B_{i,k}|^2\right)$$

i.e.

$$\|AB\|_{Fr}^2 \le \|A\|_{Fr}^2 \|B\|_{Fr}^2.$$

About differentiating series. *Let us recall that, by dominated convergence,*

$$\frac{d}{dt}\left(\sum_{k=0}^{\infty} \frac{t^k}{k!} A^k\right) = \sum_{k=0}^{\infty} \frac{d}{dt}\left(\frac{t^k}{k!} A^k\right)$$

because:

- $\sum_{k=0}^{\infty} \frac{t^k}{k!} A^k$ *is well defined for any* $t \in [0,T]$,
- $\sum_{k=0}^{\infty} \frac{d}{dt}\left(\frac{t^k}{k!} A^k\right)$ *converges uniformly in* $[0,T]$: $\sum_{k=0}^{\infty} \left\| k\frac{t^{k-1}}{k!}\|A\|_{Fr}^k \right\|_{L^\infty([0,T])} < \infty$.

(b) *Consider* X *the solution of* (1)*, define* $Y(t) := \mathrm{e}^{-tA} X(t)$*. Differentiate* Y *and show that*

$$\frac{d}{dt}\big(\mathrm{e}^{-tA} X(t)\big) = \mathrm{e}^{-tA} B(t).$$

Deduce that, for all $t \in [0,T]$,

$$X(t) \;=\; \mathrm{e}^{tA} X_0 \;+\; \int_0^t \mathrm{e}^{(t-s)A}\, B(s)\, ds.$$

Correction. *We have (since* $\frac{d}{dt} e^{-tA} = -e^{-tA}A = -Ae^{-tA}$*, using that* A *and* e^{-tA} *commutes)*

$$\frac{d}{dt}\big(e^{-tA} X(t)\big) = -Ae^{-tA}X(t) + e^{-tA}(AX(t) + B(t)) = e^{-tA}B(t).$$

Integrating:

$$e^{-tA}X(t) - X(0) = \int_0^t e^{-sA} B(s)\, ds \quad \Rightarrow \quad X(t) = e^{tA}X_0 + \int_0^t e^{(t-s)A} B(s)\, ds.$$

This is Duhamel's (variation-of-constants) formula.

(c) ***A simple example.*** *Compute* $X(t)$ *explicitly (without the integral term) in the case* $B(t) = \mathrm{e}^{-\alpha t}\mathbb{1}$*, where* $\mathbb{1} = (1, \dots, 1)^\top \in \mathbb{R}^d$*, and* $\alpha \in \mathbb{R}$*, assuming that* $\alpha \notin \sigma(-A)$ *the spectrum of* $-A$*, so that* $A + \alpha I$ *is invertible.*

Correction. *Using Duhamel,*

$$\begin{aligned} X(t) &= e^{tA}X_0 + \int_0^t e^{(t-s)A}\, e^{-\alpha s}\mathbb{1}\, ds \\ &= e^{tA}X_0 + e^{tA}\int_0^t e^{-sA}\, e^{-\alpha s}\mathbb{1}\, ds \\ &= e^{tA}X_0 + e^{tA}\left(\int_0^t e^{-s(A+\alpha I)}\, ds\right)\mathbb{1}. \end{aligned}$$

We used here the fact that for any two matrices $A, B \in \mathbb{R}^{d\times d}$*, if* $AB = BA$ *(*A *and* B *commute)*

$$e^{A+B} = e^A e^B$$

and, for $\lambda \in \mathbb{R}$

$$e^{\lambda I} = e^{\lambda} I.$$

(This can be checked using the definition of the exponantial of matrices as a series.)
If $A + \alpha I$ *is invertible, by a simple differentiation of the right hand side, one has that*

$$\int_0^t e^{-s(A+\alpha I)}\, ds = (A+\alpha I)^{-1}\Big(I - e^{-t(A+\alpha I)}\Big),$$

so

$$X(t) = e^{tA}X_0 + e^{tA}(A+\alpha I)^{-1}\Big(I - e^{-t(A+\alpha I)}\Big)\,\mathbb{1} = e^{tA}\big[X_0 + (A+\alpha I)^{-1}\mathbb{1}\big] - (A+\alpha I)^{-1}\, e^{-\alpha t}\,\mathbb{1}.$$

We used here the fact that e^{tA} *and* $(A+\alpha I)^{-1}$ *commute (since* A *and* $(A+\alpha I)^{-1}$ *commute).*

Non inversible case: *If* A *is diagonalizable and* $-\alpha$ *is an eigenvalue of* A*, then* $(A+\alpha I)$ *is not invertible. In this situation, an option is to bring* A *into Jordan (or triangular) form: the blocks corresponding to the eigenvalue* $-\alpha$ *produce additional polynomial factors in* t *multiplying the usual exponentials. Let us illustrate this in dimension 1. If* $\dot{x}(t) = -\alpha x(t) + \exp(-\alpha t)$*, then*

$$\frac{d}{dt}\left(x(t)\exp(\alpha t)\right) = \exp(\alpha t)\left(\dot{x}(t) + \alpha x(t)\right) = 1$$

which yields

$$x(t) = x(0)\exp(-\alpha t) + t\exp(-\alpha t).$$

Alternatively, another formulation is possible, indeed, one has:

$$\begin{aligned} X(t) &= e^{tA}X_0 + e^{tA}\int_0^t e^{-s(A+\alpha I)}\, ds\mathbb{1} \\ &= e^{tA}X_0 + e^{tA}\int_0^t \sum_{k=0}^{+\infty} \frac{(-1)^k s^k}{k!}(A+\alpha I)^k\, ds\mathbb{1} \\ &= e^{tA}X_0 + e^{tA}\sum_{k=0}^{+\infty} \frac{(-1)^k t^{k+1}}{(k+1)!}(A+\alpha I)^k\mathbb{1}. \end{aligned}$$

Notice that in the inversible case, our two formula are equivalent. This series form is interesting, in particular numerically, as it is how one compute the matrix exponential in pratice: by truncating this series.

Exercise 2 (About Lemma 6 (Grönwall's inequality) and applications)**.**

(a) (Grönwall) *Let* $T > 0$*. Suppose* $u \in C^0([0,T])$ *and, for all* $t \in [0,T]$*,*

$$u(t) \;\le\; \int_0^t a(s)\,u(s)\,ds \;+\; b(t),$$

with $a, b \in C^0([0,T])$ *and* $a \ge 0$*. Show that, for all* $t \in [0,T]$*,*

$$u(t) \;\le\; b(t) \;+\; \int_0^t b(s)\,a(s)\,\exp\Big(\int_s^t a(r)\,dr\Big)\,ds.$$

Hint: *Set* $v(t) := \int_0^t a(s)\,u(s)\,ds$, *and differentiate it.*

Correction. *Define* $v(t) := \int_0^t a(s)u(s)\,ds$. *Then (since* $a \geq 0$*)*

$$\dot{v}(t) = a(t)u(t) \leq a(t)\big(v(t) + b(t)\big),$$

i.e.

$$\frac{d}{dt}\Big(e^{-\int_0^t a(r)\,dr}\,v(t)\Big) = e^{-\int_0^t a(r)\,dr}\Big(\dot{v}(t) - a(t)v(t)\Big) \leq\ e^{-\int_0^t a(r)\,dr}\,a(t)\,b(t).$$

Integrating from 0 *to* t *(with* $v(0) = 0$*) yields*

$$v(t)\ \leq\ \int_0^t a(s)\,b(s)\,\exp\Big(\int_s^t a(r)\,dr\Big)\,ds.$$

Since $u(t) \leq v(t) + b(t)$, *the claimed bound follows.*

(b) (Continuous dependence / perturbation estimate) *Let* $f : \mathbb{R}^d \to \mathbb{R}^d$ *be continuous and globally* L*-Lipschitz. Let* $x : [0,T] \to \mathbb{R}^d$ *be a* C^1 *function such that*

$$\begin{cases} \dot{x} = f(x), & \text{on } [0,T], \\ x(0) = x_0, \end{cases}$$

and let $y : [0,T] \to \mathbb{R}^d$ *be a* C^1 *function satisfying the perturbed problem*

$$\begin{cases} \dot{y} = f(y) + \varepsilon, & \text{on } [0,T], \\ y(0) = y_0, \end{cases}$$

where $\varepsilon : [0,T] \to \mathbb{R}^d$ *is continuous. Using question (a), show that for all* $t \in [0,T]$,

$$\|x(t) - y(t)\|\ \leq\ \mathrm{e}^{LT}\left(\|x_0 - y_0\| + \int_0^T \|\varepsilon(s)\|\,ds\right).$$

Hint: *write an integral equation for* $w := x - y$ *and apply question (a) with* $u(t) = \|w(t)\|$.

Correction. *Set* $w(t) := x(t) - y(t)$. *Integrating the two equations between* 0 *and* t *and subtracting,*

$$w(t) = x_0 - y_0 + \int_0^t \big(f(x(s)) - f(y(s)) - \varepsilon(s)\big)\,ds.$$

Taking norms and using that f *is globally* L*-Lipschitz,* $\|f(x(s)) - f(y(s))\| \leq L\|w(s)\|$, *so*

$$\|w(t)\|\ \leq\ \|x_0 - y_0\| + \int_0^t \|\varepsilon(s)\|\,ds + \int_0^t L\,\|w(s)\|\,ds\ \leq\ C + \int_0^t L\,\|w(s)\|\,ds,$$

where we set the constant

$$C := \|x_0 - y_0\| + \int_0^T \|\varepsilon(s)\|\,ds\,,$$

using $\int_0^t \|\varepsilon(s)\|\,ds \leq \int_0^T \|\varepsilon(s)\|\,ds$ *since the integrand is nonnegative.*
We now apply question (a) with

$$u(t) = \|w(t)\|, \qquad a(s) \equiv L \geq 0, \qquad b(t) \equiv C.$$

The hypothesis $u(t) \le \int_0^t a(s)u(s)\,ds + b(t)$ is exactly the inequality above, hence

$$\|w(t)\| \;\le\; C + \int_0^t C\,L\,\exp\Big(\int_s^t L\,dr\Big)\,ds = C + C\int_0^t L\,\mathrm{e}^{L(t-s)}\,ds = C + C\big(\mathrm{e}^{Lt}-1\big) = C\,\mathrm{e}^{Lt}.$$

Since $t \le T$ and $C \ge 0$, we conclude

$$\|x(t)-y(t)\| \;\le\; C\,\mathrm{e}^{Lt} \;\le\; \mathrm{e}^{LT}\left(\|x_0-y_0\| + \int_0^T \|\varepsilon(s)\|\,ds\right).$$

Comments. *Taking $\varepsilon \equiv 0$ gives the continuous dependence of the solution on the initial datum (and, in particular, uniqueness for the Cauchy problem associated with a globally Lipschitz f). The estimate is also the basic building block for the convergence analysis of one-step numerical schemes, where ε plays the role of the (local) consistency error.*

Exercise 3 (Non-uniqueness when f is not Lipschitz: $\dot{x} = \sqrt{|x|}$)**.** *Consider the ordinary differential equation:*

$$\forall t \ge 0,\ \dot{x}(t) = \sqrt{|x(t)|}, \qquad x(0) = 0.$$

(a) *Exhibit two distinct solutions.*

Correction. *One solution is the trivial one: $x(t) \equiv 0$ for all $t \ge 0$ (since $\sqrt{|0|} = 0$). A second (nontrivial) solution can be found by separation of variables on the set $\{x \ge 0\}$:*

$$\frac{dx}{dt} = \sqrt{x} \quad\Longrightarrow\quad \frac{dx}{\sqrt{x}} = dt \quad\Longrightarrow\quad 2\sqrt{x} = t + C.$$

Imposing $x(0) = 0$ gives $C = 0$ and thus

$$x(t) = \frac{t^2}{4}, \qquad t \ge 0.$$

Both $x = 0$ and $x(t) = t^2/4$ satisfy the ODE, so uniqueness fails.

(b) *Explain why this does not contradict the Cauchy-Lipschitz (Picard–Lindelöf) theorem by proving that $f(x) = \sqrt{|x|}$ is* not *locally Lipschitz at $x = 0$.*

Correction. *The Cauchy-Lipschitz (Picard–Lindelöf) theorem requires f to be (locally) Lipschitz in a neighborhood of the initial value. Here $f(x) = \sqrt{|x|}$ is not locally Lipschitz at 0. Indeed, if there were $L > 0$ and $\delta > 0$ such that $|\sqrt{|x|} - \sqrt{|y|}| \le L|x-y|$ for all $|x|, |y| < \delta$, then taking $y = 0$ gives*

$$\sqrt{|x|} \le L|x| \quad \text{for } |x| < \delta,$$

i.e. $\frac{\sqrt{|x|}}{|x|} \le L$, but $\frac{\sqrt{|x|}}{|x|} = |x|^{-1/2} \to \infty$ as $x \to 0$, a contradiction. Thus f is not (locally) Lipschitz at 0.

(c) *Show that for any $t_0 \ge 0$,*

$$x_c(t) := \begin{cases} 0, & 0 \le t \le t_0, \\ \dfrac{(t-t_0)^2}{4}, & t \ge t_0, \end{cases}$$

is a solution; conclude that there are infinitely many solutions.

Correction. *For any $t_0 \geq 0$, define*

$$x(t) = \begin{cases} 0, & t \leq t_0, \\ \dfrac{(t-t_0)^2}{4}, & t \geq t_0. \end{cases}$$

Then x is C^1, satisfies $x(0) = 0$, and for $t < t_0$ we have $\dot{x} = 0 = \sqrt{|0|}$; for $t > t_0$ we have $\dot{x}(t) = \frac{t-t_0}{2} = \sqrt{(t-t_0)^2/4} = \sqrt{|x(t)|}$. At $t = t_0$, both sides are 0. Hence x is a solution. Taking different t_0 yields infinitely many distinct solutions.

Exercise 4. *The logistic equation*

$$\dot{x}(t) \;=\; r\,x(t)\Big(1 - \frac{x(t)}{K}\Big) \tag{2}$$

models the evolution of the density $x(t)$ of a species depending on a limited resource, e.g., sheep on an island, with no predators, but needing to find grazing fields. When the density of the population is low, the per-capita growth rate is close to $r > 0$. When the density increases, the growth rate decreases. It becomes negative if the density is higher than the carrying capacity K of the environment. This equation can be solved explicitly, e.g., by separation of variables, but the aim of this exercise is to show that its qualitative behavior may be understood without having an explicit formula for x.

Let $x_0 \in \mathbb{R}_+$. Let $(J, x(\cdot))$ denote the maximal solution of (2) such that $x(0) = x_0$.

1. *What happens if $x_0 = 0$ or $x_0 = K$?*

   Correction.

   If $x_0 = 0$ (resp. $x_0 = K$), then $x(t) = 0$ (resp. $x(t) = K$) for all $t \in J = \mathbb{R}$ is a solution. Then by Cauchy-Lipschitz theorem, this is the unique solution.

2. *Assume $0 < x_0 < K$.*

   (a) *Show that $0 < x(t) < K$ for all $t \in J$ and that $J = \mathbb{R}$.*
   Show that $x(\cdot)$ is strictly increasing.

   (b) *Show that $x(t)$ has a limit in $[x_0, K]$ as $t \to +\infty$ and in $[0, x_0]$ as $t \to -\infty$.*

   (c) *Show that if $x(t) \to x^* \in \mathbb{R}$ as $t \to +\infty$, then $f(x^*) = 0$, where $f(x) = r\,x\big(1 - \frac{x}{K}\big)$.*

   (d) *Show that $x(t) \xrightarrow[t\to+\infty]{} K$ and $x(t) \xrightarrow[t\to-\infty]{} 0$.*

   Correction.

   (a) *By contradiction, we assume there exists $t^* \in J$ such that $x(t^*) \leq 0$ (resp. $x(t^*) \geq K$). By the intermediate value theorem, there exists at least one $t \in [0, t^*]$ (or $t \in [t^*, 0]$ if $t^* < 0$) such that $x(t) = 0$ (resp. $x(t) = K$). But this implies by uniqueness of the solution to (2) (notice that $x \mapsto rx(1 - x/K)$ is locally Lipschitz) that for all time $x_0 = 0$ (resp. $x_0 = K$) which leads to a contradiction. For all $t \in J$, $0 < x(t) < K$. Since $x(\cdot)$ is bounded, the maximal solution is global by the Theorem 8 from the lecture notes: $J = \mathbb{R}$.*
   Finally, for all $t \in J, 1 - x(t)/K > 0$ and $x(t) > 0$, so $\dot{x}(t) > 0$ for all $t \in J$ and $x(\cdot)$ is strictly increasing.

(b) *Since $x(\cdot)$ is bounded, it admits a finite supremum. Let's call it L. By definition, for all $t \in \mathbb{R}$, $x(t) \leq L$. As L is the supremum of $x(\cdot)$, for all $\varepsilon > 0$, there exists $t_\varepsilon \in \mathbb{R}$ such that $|x(t_\varepsilon) - L| < \varepsilon$. $x(\cdot)$ being strictly increasing, for all $t > t_\varepsilon, |x(t) - L| < \varepsilon$. Thus, L is indeed the limit of $x(\cdot)$ as t goes to infinity. The same proof can be done with L being the infimum of $x(\cdot)$ in order to answer the second part of the question.*

(c) *Fix $h > 0$. By the Mean Value Theorem applied to the function $s \mapsto x(s)$ on the interval $[t, t+h]$, there exists a point $c_t \in (t, t+h)$ such that*

$$\frac{x(t+h) - x(t)}{h} = \dot{x}(c_t) = f(x(c_t)).$$

Since $x(t) \to x^*$ as $t \to +\infty$, we also have $x(c_t) \to x^*$. By continuity of f,

$$\lim_{t \to +\infty} f(x(c_t)) = f(x^*).$$

On the other hand, because $x(t) \to x^*$,

$$\lim_{t \to +\infty} \frac{x(t+h) - x(t)}{h} = \frac{x^* - x^*}{h} = 0.$$

Therefore, by comparing the two limits,

$$f(x^*) = 0.$$

(d) *The answer to the previous question can be adapted to show that if $x(t) \to x_* \in \mathbb{R}$ as $t \to -\infty$, then $f(x_*) = 0$ as well. Since $0 < x(t) < K$ for all $t \in \mathbb{R}$ and $x(\cdot)$ is strictly increasing, and since the two unique roots of f are 0 and K, we have the result.*

3. *Assume $x_0 > K$. Show that $x(t) > K$ for all $t \in J$, that $x(\cdot)$ is decreasing, that $\sup J = +\infty$ and that $x(t) \to K$ as $t \to +\infty$.*
Show that $x(t) \to +\infty$ as $t \to \inf J$.

Correction. *Similarly as in question 2 (a), the intermediate value theorem and Cauchy Lipschitz theorem implie that $x(t) > K$ for all $t \in J$. Hence for all $t \in J$, $x(t) > 0$ and $1 - x(t)/K < 0$ so $\dot{x}(t) < 0$ for all $t \in J$ and then $x(\cdot)$ is strictly decreasing on J. As $x(\cdot)$ is bounded from below and decreasing, it is bounded on $\mathbb{R}^+$ so by non-blowup criterion (Theorem 8), $\sup J = +\infty$. Using the same argument as in question 2 (c), we have that $x(t) \to K$ as $t \to +\infty$.*

4. *Assume $x_0 < 0$ (a case with limited ecological interest). Show that $x(t) < 0$ for all $t \in J$, that $x(\cdot)$ is decreasing, that $\inf J = -\infty$ and $x(t) \to 0$ as $t \to -\infty$.*
Show that $x(t) \to -\infty$ as $t \to \sup J$.

Correction. *The questions 4 can be answered using same arguments as in questions 3.*

5. *Sketch the graph of representative solutions and the phase line.*

Correction. *See figure 1 ($r = K = 1$).*

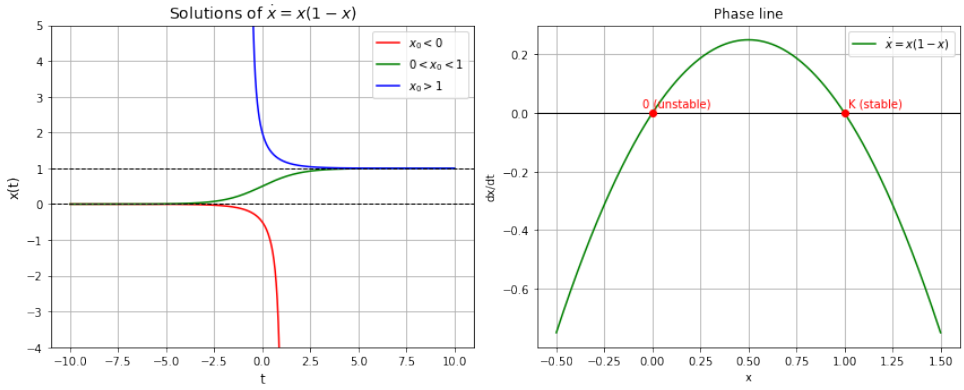


Figure 1: Left: solutions for $x_0 = -0.5$, $x_0 = 0.5$ and $x_0 = 2$. Right: phase line.

Exercise 5 (Consistency of some ODE solvers)**.** *Consider the ordinary differential equation*

$$\dot{x}(t) = f(x(t)),$$

where $f : \mathbb{R} \to \mathbb{R}$ is sufficiently smooth. Let $\Delta t > 0$ and $t_n = n\Delta t$. We consider the two explicit one-step schemes

- Explicit Euler*:*
$$x_{n+1} = x_n + \Delta t\, f(x_n)$$
- Heun's method*:*
$$x_{n+1} = x_n + \frac{\Delta t}{2}\big(f(x_n) + f(x_n + \Delta t\, f(x_n))\big)$$

(a) *Verify mathematically that explicit Euler is of order 1 and that Heun's method is of order 2.*

Hint*: Use Taylor expansions to evaluate the difference at time t_{n+1} between the exact solution $x(t_{n+1})$ and the numerical solution x_{n+1}, assuming $x(t_n) = x_n$.*

Correction.

Taylor expansion of the exact solution

Assuming that $x(t_n) = x_n$ the exact solution satisfies:

$$x(t_{n+1}) = x_n + \Delta t x_n' + \frac{\Delta t^2}{2} x_n'' + \frac{\Delta t^3}{6} x_n''' + \frac{\Delta t^4}{24} x^{(4)}(\xi_n), \quad \xi_n \in [t_n, t_{n+1}]$$

where (since $\dot{x}(t) = f(x(t))$):

$$x_n' = \dot{x}(t_n) = f(x_n), \quad x_n'' = \ddot{x}(t_n) = f'(x_n)f(x_n), \quad x_n''' = \dddot{x}(t_n) = f''(x_n)f(x_n)^2 + f'(x_n)^2 f(x_n).$$

Explicit Euler. *The scheme is $x_{n+1} = x_n + \Delta t\, f(x_n)$, so subtracting,*

$$x(t_{n+1}) - x_{n+1} = \frac{\Delta t^2}{2} f'(x_n)f(x_n) + \mathcal{O}(\Delta t^3) = \mathcal{O}(\Delta t^2).$$

The leading coefficient is $\frac{1}{2}f'(x_n)f(x_n)$, so the one-step error is exactly of order Δt^2 and explicit Euler is of order 1.

Heun's method

$$x_{n+1} = x_n + \frac{\Delta t}{2}\big(f(x_n) + f(x_n + \Delta t f(x_n))\big).$$

Expand $f(x_n + \Delta t f(x_n))$ in Taylor series:

$$\begin{aligned} f(x_n + \Delta t f(x_n)) &= f(x_n) + \Delta t f'(x_n) f(x_n) + \frac{\Delta t^2}{2} f''(x_n) f(x_n)^2 + \mathcal{O}(\Delta t^3) \\ &= f(x_n) + \Delta t x_n'' + \frac{\Delta t^2}{2} f''(x_n) f(x_n)^2 + \mathcal{O}(\Delta t^3). \end{aligned}$$

Hence

$$x_{n+1} = x_n + \Delta t f(x_n) + \frac{\Delta t^2}{2} x_n'' + \frac{\Delta t^3}{4} f''(x_n) f(x_n)^2 + \mathcal{O}(\Delta t^4).$$

By comparing with the exact solution, $|x(t_{n+1}) - x_{n+1}| = \mathcal{O}(\Delta t^3)$, which yields on order 2 rate of convergence.

(b) *Verify numerically the orders found in (a). Pick a test problem with a known exact solution, e.g. $f(x) = \lambda x$ (so $x(t) = x_0 e^{\lambda t}$) or the logistic right-hand side. On a fixed interval $[0, T]$, and for a sequence of decreasing time steps Δt, compute the error*

$$E(\Delta t) := \max_{0 \le n \le N} \big|x(t_n) - x_n\big| \qquad (\textit{or simply } |x(T) - x_N|),$$

and plot $E(\Delta t)$ against Δt in log–log scale. Check that the slopes are close to 1 for explicit Euler and to 2 for Heun's method.

Correction. *In log–log scale one obtains essentially straight lines of slope ≈ 1 for explicit Euler and ≈ 2 for Heun's method, in agreement with (a). With $T = 1$, $\dot{x} = x$:*

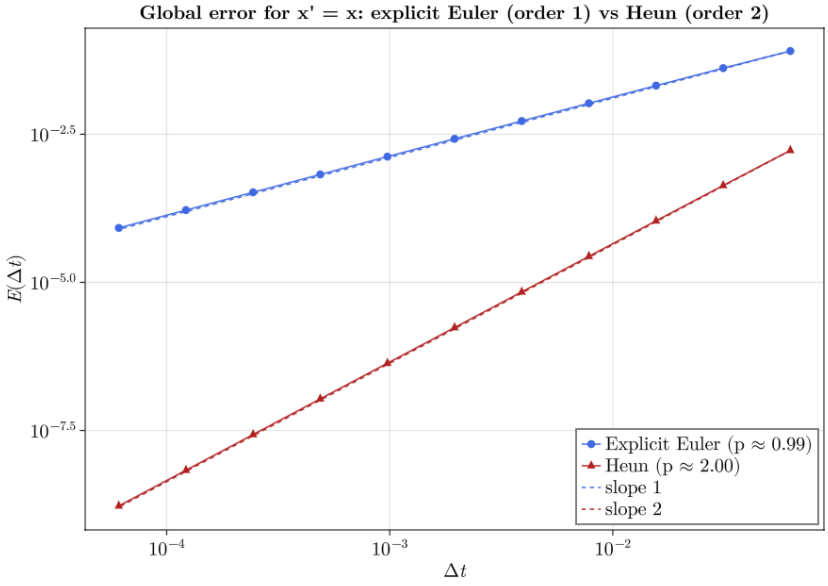


Figure 2: Log-log plots of the numerical error for explicit Euler and Heun's methods

Exercise 6 (A stiff model: explicit vs. implicit Euler)**.** *Consider the scalar ODE*

$$\dot{x}(t) = f(t,x) := \alpha\big(x(t) - \sin t\big) + \cos t, \qquad x(0) = x_0, \tag{3}$$

with a constant parameter $\alpha \in \mathbb{R}$*.*

(a) The analytical solution. *Find an explicit solution to* (3) *(Duhamel formula or any other technique can be used). Identify the long-time behavior of solutions depending on the sign of* α*.*

Correction. *Let us introduce* $y(t) := x(t) - \sin t$ *gives* $y'(t) = x'(t) - \cos t = \alpha(x(t) - \sin t) = \alpha y(t)$*, so*

$$x(t) = \sin t + x_0\, e^{\alpha t}.$$

Hence:

- *if* $\alpha < 0$*,* $x(t) \to \sin t$ *exponentially fast on time scale* $|\alpha|^{-1}$*;*
- *if* $\alpha > 0$*, the exponential term dominates and solutions blow up exponentially (unless* $x_0 = 0$*).*

In the following, we assume $\alpha < 0$*.*

(b) Time discretization (explicit/implicit Euler). *Let* $t_n = n\Delta t$*. Derive (we use the notation* $x_n \approx x(t_n)$*):*

- *the* explicit Euler *scheme;*
- *the* implicit Euler *scheme; solved in closed form for* x_{n+1}*.*

Correction. Explicit Euler:

$$x_{n+1} = x_n + \Delta t\, f(t_n, x_n) = x_n + \Delta t\big[\alpha(x_n - \sin t_n) + \cos t_n\big].$$

Implicit Euler:

$$x_{n+1} = x_n + \Delta t\, f(t_{n+1}, x_{n+1}) = x_n + \Delta t\big[\alpha(x_{n+1} - \sin t_{n+1}) + \cos t_{n+1}\big].$$

Collecting x_{n+1}*-terms,*

$$(1 - \alpha\Delta t)\, x_{n+1} = x_n + \Delta t\big[-\alpha \sin t_{n+1} + \cos t_{n+1}\big],$$

so (provided $1 - \alpha\Delta t \neq 0$*)*

$$x_{n+1} = \frac{x_n + \Delta t\big[-\alpha \sin t_{n+1} + \cos t_{n+1}\big]}{1 - \alpha\Delta t}.$$

Why is this model stiff*? One says that a problem is stiff if the stability constraint is more stringent to determine the time step than the precision constraint. This typically occurs in systems with two time scales: on the fast timescale, the system relaxes towards the dynamics with slow time scale. In our context, the fast time scale is related to the parameter* α*. On our example, for* $\alpha < 0$*, solutions relax to* $\sin t$ *with fast time scale* $|\alpha|^{-1}$*. When* $|\alpha| \gg 1$ *there are two widely separated time scales:*

$$\textit{fast} = |\alpha|^{-1}, \qquad \textit{slow/forcing} = 2\pi,$$

so the problem is stiff*.*

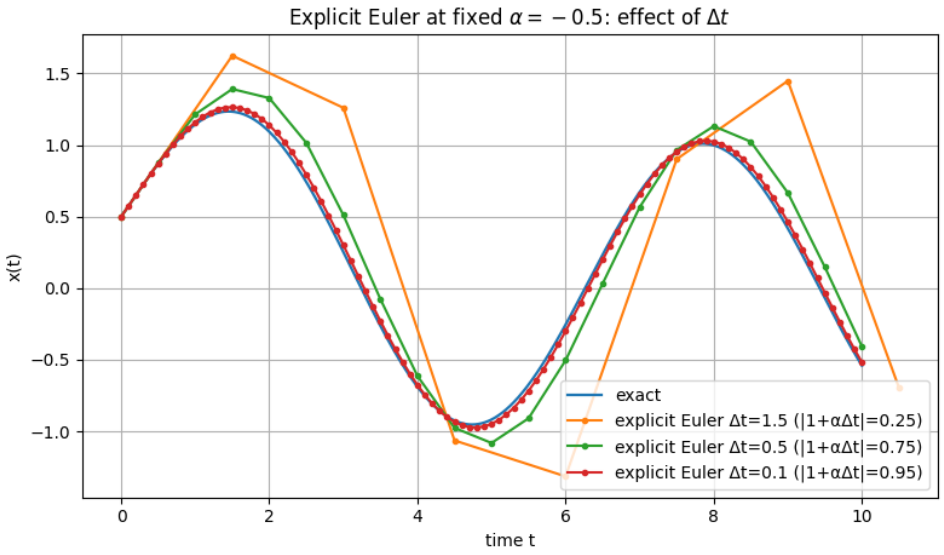


Figure 3: Solution of explicit Euler for different values of Δt. We see the influence of the time step on the quality of the solution. The larger the time step is, the further the numerical approximation is from the exact solution

(c) *Simulations.* *Implement both schemes to compute an approximation over the time interval* $[0, 10]$ *and run them for* $\alpha = -0.5$. *For the explicit scheme, vary* Δt *to see its influence.*

Correction. *See Figure 3.*

(d) *Absolute stability insight.* *Use different values of* $\alpha < 0$ *and* Δt *to illustrate the stiffness of the problem. For instance* $(\alpha, \Delta t) = (-5, 0.01), (-10, 0.18), (-10, 0.201), (-10, 0.22)$. *Show numerically that explicit Euler is stable only if* $|1 + \alpha\Delta t| < 1$, *whereas implicit Euler is always stable for* $\alpha < 0$.

Correction. *See Figure 4. Why do we consider the condition* $|1 + \alpha\Delta t| < 1$*? An easy understanding is to look at the homogeneous part of the ODE, i.e.* $\dot{x} = \alpha x$. *Discretised with explicit Euler, it reads* $x_{n+1} = (1 + \alpha\Delta t)x_n$. *Thus*

$$\left|\frac{x_{n+1}}{x_n}\right| = |1 + \alpha\Delta t|.$$

This sequence x_n *goes to zero only if* $|1 + \alpha\Delta t| < 1$.

(e) *Order of accuracy.* *Choose a time step small enough that the schemes are stable in your setting. Verify numerically that both explicit and implicit Euler are first-order accurate in* Δt.

Correction. *See Figure 5.*

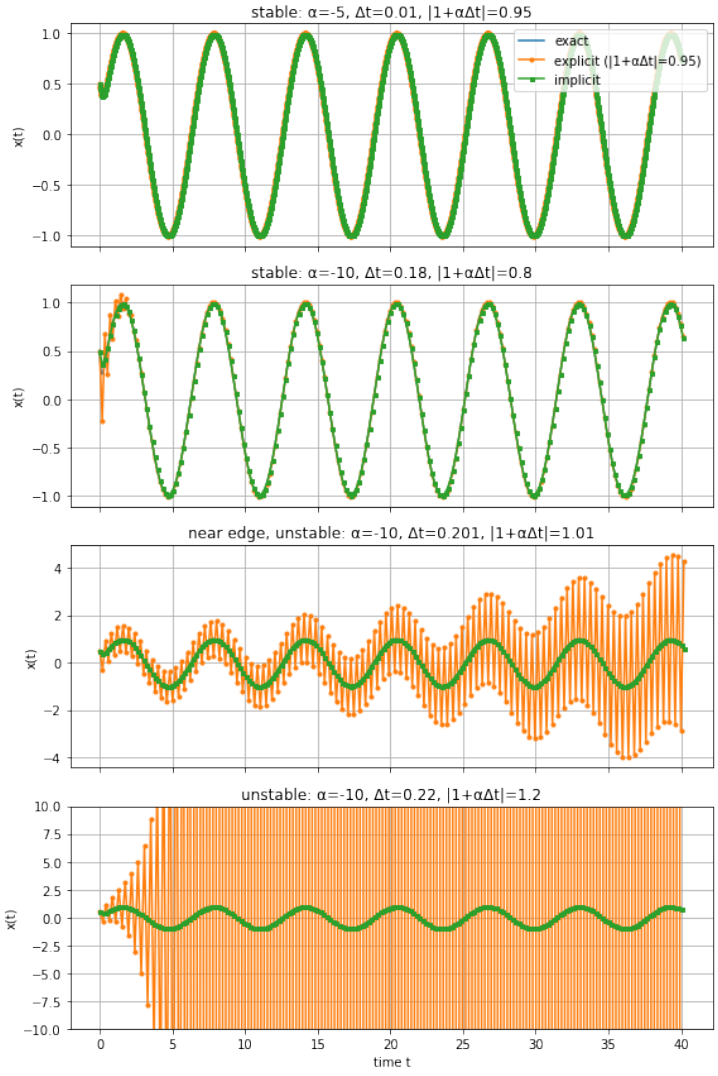


Figure 4: Implicit Euler always give stable solution, however it is not the case for explicit Euler. When the constant $|1 + \alpha\Delta t| > 1$ (third and fourth plots) the numerical scheme fails due to the stiffness of the model. We can also see consequences of this stiffness on the second plot, but as $|1 + \alpha\Delta t| < 1$, the perturbations around $t = 0$ are damped.

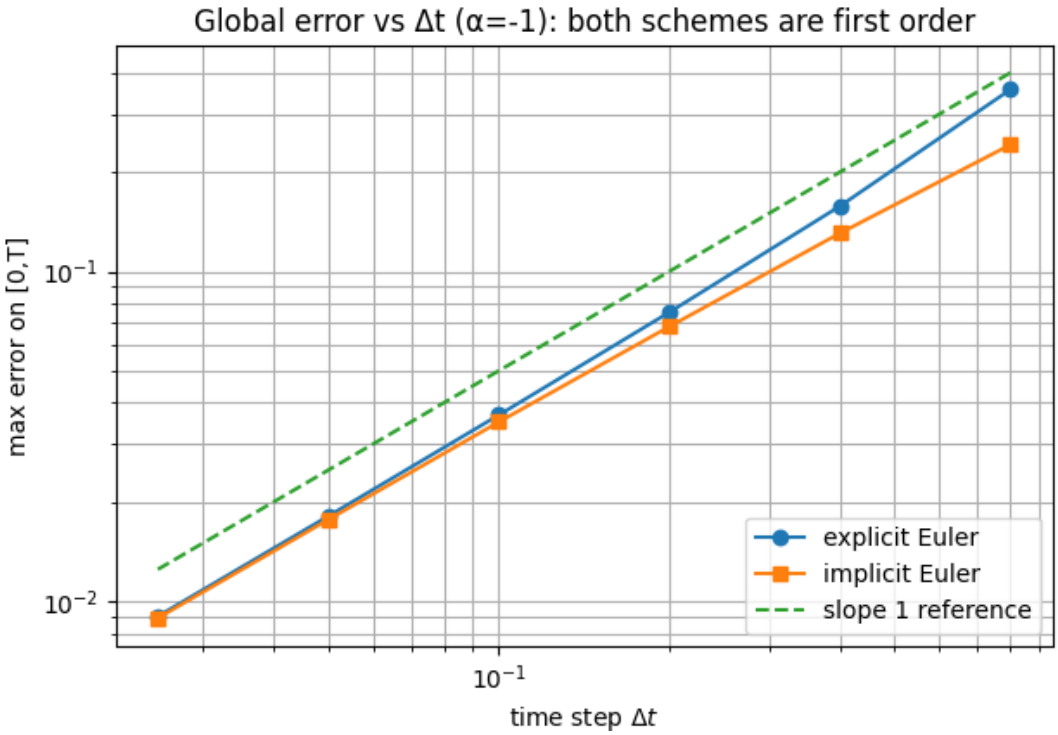


Figure 5: We find the expected first order scaling for both methods.